MIL-STD-1647C
20 December 1987
SUPERSEDING
MIL-STD-1647B
21 October 1983

MILITARY STANDARD

IDENTIFICATION MARKINGS FOR DOMESTICALLY MANUFACTURED BEARINGS, BALL, ANNULAR FOR INSTRUMENTS AND PRECISION COMPONENTS

AMSC N/A

FSC 3110

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DEPARTMENT OF DEFENSE
Washington, DC 20301

Identification Markings for Domestically Manufactured Bearings, Ball, Annular for Instruments and Precision Components.

1. This military standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (Code 53), Lakehurst, NJ 08733-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. SCOPE

1.1 Scope. This standard specifies the marking requirements for identification and certification of miniature and instrument type annular ball bearings of domestic manufacture.

1.2 Purpose. This standard establishes the literal and symbolic markings to be used for the identification of bearings manufactured in the United States and Canada for use in defense contracts. The markings chosen comply with those established and accepted as standard by the bearing industry.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-81793 - Bearings, Ball, Annular, for Instruments and Precision Rotating Components.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-1334 - Process for Barrier Coating of Anti-friction Bearings.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this standard to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

American Iron and Steel Institute (AISI)

AISI 440C - Corrosion-resistant Steel

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 16th Street NW, Washington, DC 20036.)

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Aerospace Materials Specifications for Flight Vehicle Construction (AMS)

AMS 6444 - Steel Bars, Forgings, and Tubing 1.45CR (0.98 - 1.10C) (SAE 52100) Premium Aircraft Quality, Consumable Electrode Vacuum Melted.

Society of Automotive Engineers (SAE)

SAE 52100 - Chrome Steel

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.

3. DEFINITIONS

3.1 Miniature and instrument type annular ball bearings. For the purpose of this standard, any reference to "bearings" or "ball bearings" shall be understood to mean miniature and instrument type annular ball bearings as defined in MIL-B-81793.

3.2 Domestic manufacture. Domestically manufactured ball bearings are those which are assembled in the United States or Canada using components all of which have been manufactured completely in the United States or Canada. When a ball bearing assembly is involved, all components of the assembly must also have been manufactured in the United States or Canada.

3.3 Made in U.S.A.. Made in U.S.A. refers to ball bearings and components of ball bearings completely manufactured, assembled and tested within the United States of America.

3.4 Basic bearing part number. The basic bearing part number refers to the group of numbers or letters in a manufacturer's part number which specifies the bore, outside diameter, width, material and the number of shields or seals used. Numbers and letter codes may be as defined in MIL-B-81793.

4. GENERAL REQUIREMENTS

4.1 Identification. All bearings used in any defense contract, or bearings supplied for the repair of a defense contract item, shall be marked in accordance with one of the following methods:

4.1.1 Literal (non-symbolic) identification. The manufacturer's name, basic bearing part number and "Made in U.S.A.", "U.S.A." or "Made in Canada", "Canada" shall be marked on bearings one inch (25.4 mm) or greater in outside diameter having ring face width of .0625 inch (1.575 mm) or greater. The markings shall be on the face of the inner or outer ring to assure that the basic bearing can still be fully identified when the shields or seals are removed.

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4.1.2 Symbolic marking. The applicable symbolic marking illustrated in Figure 1 shall be marked on the inner or outer ring face of bearings with an outside diameter less than one inch (25.4 mm) or where the usable face width is less than .0625 inch (1.575 mm). Symbolic markings shall fully identify the manufacturer of the bearing and the material used in its manufacture (AISI 440C or SAE 52100). They shall also indicate that the bearing was manufactured to tolerances which are equal to or exceed the requirements for ABEC 5P of MIL-B-81793.

4.1.3 Indirect marking. Bearings having an inadequate face width for the symbolic markings of 4.1.2 (Figure 1 symbolic markings) shall be exempt from the marking requirements of 4.1.1 and 4.1.2 The unit packages of such bearings shall be marked with the information specified in 4.1.1 of this standard and the method of such marking shall be in accordance with MIL-STD-129.

4.1.4 Barrier coated bearing marking. In addition to the requirements of 4.1.1, 4.1.2 and 4.1.3, barrier coated bearings shall be marked in accordance with MIL-STD-1334. Such markings shall be accomplished prior to barrier coating of the bearing.

4.1.5 Duplex bearing marking. Duplex bearings shall be marked in accordance with the requirements of MIL-B-81793.

4.1.6 Exemptions from marking requirements. Bearings made of materials other than AISI 440C and SAE 52100 steels, bearings of configuration other than conventional radial or angular contact type, and bearings where marking is specifically prohibited by the Government shall be exempt from all marking requirements of this standard. Such bearings shall be marked per 4.1.1 through 4.1.5 only when specified in the applicable contract (except where marking is prohibited).

4.2 Certification of symbolic markings. All bearing manufacturers utilizing symbolic markings shall furnish to: Commander, Naval Air Systems Command (AIR-5143), Washington, DC 20361, a notarized certification that the use of symbolic markings shall have the same meaning as the words "Made in U.S.A.", "U.S.A." or "Made in Canada", "Canada" and shall indicate the bearing was manufactured of corrosion-resistant steel (AISI 440C) or chrome steel (SAE 52100) to tolerances which are equal to or exceed the requirements for ABEC 5P of MIL-B-81793.

4.3 Marking methods. Markings shall not result in raised metal or in any way affect the function of the bearing. The marking may be applied by stamping, electroetch, or any other permanent method. Selection of the marking method is left to the manufacturer.

5. NOTES

5.1 Supersession data. This standard supersedes MIL-STD-1647B.

5.2 Subject term (key word) listing.

Domestic manufacture
Marking, barrier coated bearing
Marking, duplex bearing

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Marking, literal
Marking, symbolic
Miniature and instrument ball bearings

5.3 Changes from previous issue. Vertical lines or astericks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - AS
Air Force - 99

Review activities:
Air Force - 84
DISC - ESA

Preparing activity:
Navy - AS

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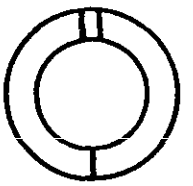
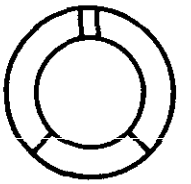
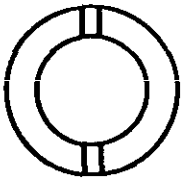
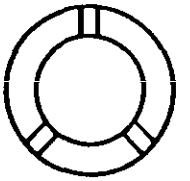
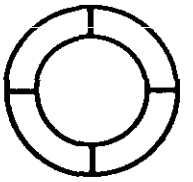
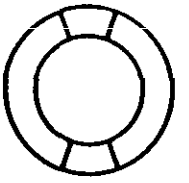
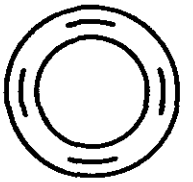
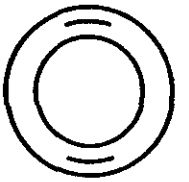
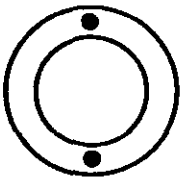
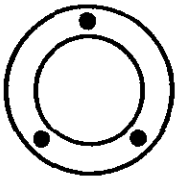
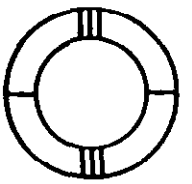
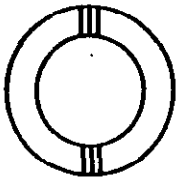
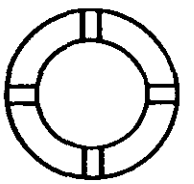
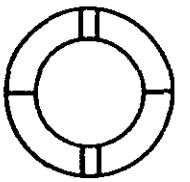
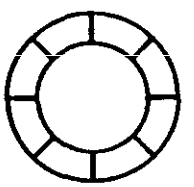
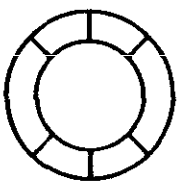
COMPANY NAME	ABEC 5P OR BETTER	
	SAE 52100	AISI 440C
FAG BEARINGS LTD.		
NEW HAMPSHIRE BALL BEARINGS (PETERBOROUGH, NH)		
MINIATURE PRECISION BEARING		
BARDEN		
FAFNIR		
MARLIN ROCKWELL CORPORATION		
NEW HAMPSHIRE BALL BEARINGS (CHATSWORTH, CA)		
PACAMOR BEARINGS INC.		

FIGURE 1. Symbolic markings.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-1647C REV C	IDENTIFICATION MARKINGS FOR DOMESTICALLY MANUFACTURED BEARINGS, BALL, ANNULAR FOR INSTRUMENTS AND PRECISION COMPONENTS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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